BCA (Sem-III) – Math

(305) GE-3

2022

Time : 1:30 hours

Full Marks : 40

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer **all** sections as directed.*

Section–A

(Compulsory)

1. Pick up the correct alternative for each of the following questions : $1\times10=10$

   (a) In tossing two coin at a time, the probability of geting at most one head is :

(i) $\frac{1}{4}$

(ii) $\frac{1}{2}$

(iii) $\frac{3}{4}$

(iv) None of these

(b) The probability that a leap year selected at random will have 53 Sunday is :

(i) $\frac{1}{7}$

(ii) $\frac{2}{7}$

(iii) $\frac{3}{7}$

(iv) None of these

(c) The probability of an impossible event is :

(i) 0

(ii) 1

(iii) $\frac{1}{2}$

(iv) ∞

(d) Two dice are thrown simultaneously. The probability of the sum of the two results is 10 is :

(i) $\frac{1}{12}$

(ii) $\frac{1}{6}$

(iii) $\frac{1}{3}$

(iv) None of these

(e) Arithmetic means of 15, 20, 25, 12 is :

(i) 16

(ii) 17

(iii) 18

(iv) None of these

(f) The sum of the deviations of a set of observed values from their Arithmetic Mean (AM) :

(i) $\overline{x}$

(ii) 2

(iii) 0

(iv) 1

(g) Relation between mean, median and mode is :

(i) Mean+Mode=Median

(ii) Mode+2Mean=3Median

(iii) Mean+2Mode=3Median

(iv) None of these

(h) The sum of the probabilities of the random event A and its complementary event A' is equal to :

(i) 0

(ii) $\frac{1}{2}$

(iii) 1

(iv) None of these

(i) The mode of the samples

2, 1, 3, 2, 5, 3, 2, 4, 2, 1, 6 is :

(i) 1

(ii) 2

(iii) 3

(iv) 4

(j) The value of correlation ratio varies from :

(i) 0 to 1

(ii) -1 *to* 0

(iii) -1 *to* 1

(iv) $-\infty$ *to* ∞

Section–B

(Short answer type questions)

Answer any **three** questions of the following :

5×3=15

2. What do you understand by measure of central tendency ?

3. Define geometric mean with properties.

4. How will you calculate median in case of ungrouped data ?

5. Discuss about Poisson distribution.

6. Define regression coefficient.

7. What do you understand by correlation between two variables ?

Section–C

(Long answer type questions)

Answer any **two** questions of the following :

7.5×2=30

8. A box contains 3 red, 4 white and 5 black balls. Two balls are drawn at random from the box. Find the probability that both the balls are red or white.

9. The following table gives the diastolic blood pressure of 250 men. The readings were made to the nearest millimeter and the central value of each group is given by :

Blood pressure (mm)	60	65	70	75	80	85	90	95
No. of men	4	5	31	39	114	30	25	2

Calculate the mean and median from the data.

10. Calculate the correlation coefficient for the following heights (in inches) of father (X) and their son (Y)

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

11. Define correlation coefficient and how it differs from regression.

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